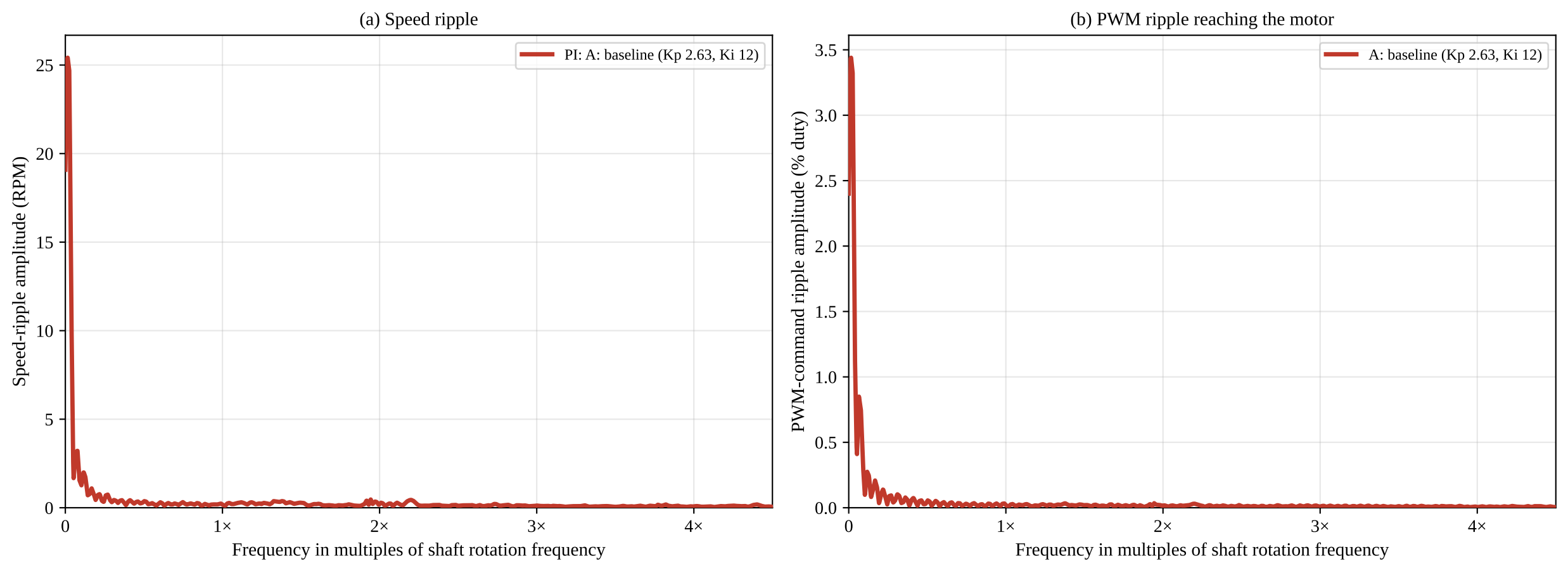
(a) Speed ripple
PI: A: baseline (Kp 2.63, Ki 12)
Speed-ripple amplitude (RPM)
0
5
10
15
20
25
0
1×
2×
3×
4×
Frequency in multiples of shaft rotation frequency
(b) PWM ripple reaching the motor
A: baseline (Kp 2.63, Ki 12)
PWM-command ripple amplitude (% duty)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0
1×
2×
3×
4×
Frequency in multiples of shaft rotation frequency